

XINYU LI

7324066689 ◇ lixinyu.arthur@outlook.com ◇ <https://www.arthurlxy.com>

EXPERIENCE

AGI, Amazon

Principal Applied Scientist

2022 – Present

Bellevue, WA

- **MM RAG / Nova Embedding:** Led the design and development of the multimodal embedding model (Nova Embedding), achieving SoTA performance across visual, document, acoustic, and text modalities, and powering production use cases across S3, Photos, and Media.
- **M-LLM / Nova:** Tech-lead for multi-modal understanding in Nova, covering video, image, acoustic, and interleaved cross-modal content. Nova2-omni models achieved strong video and multi-modal understanding across multiple benchmarks and support customers across domains.
- **Training & Ablation Pipeline:** Core contributor to the MLLM training and ablation pipeline, with full 3D-parallelism support, enabling rapid science iteration, flexible experimentation, and reproducible research across tasks and modalities.
- **Research Leadership:** Led research efforts in video understanding, multi-modal reasoning, and long-context multi-modal modeling, and drove university collaborations resulting in publications in CVPR, ICCV, ECCV, NeurIPS, and ICLR.

AML, Tiktok/ByteDance

Staff Research Scientist

2021 - 2022

Bellevue, WA

- **MM Understanding:** Led the development of a large-scale multimodal model for vision, audio, and language reasoning, supporting compliance, search, and recommendation systems.
- **Model Zoo Framework:** Developed a unified framework for implementing and benchmarking state-of-the-art vision and multimodal models, accelerating experimentation and model development cycles.
- **Research:** Led multimodal and video understanding research, with publications in CVPR, ICASSP, and ICLR.

AWS AI, Amazon

Senior Applied Scientist

2018 - 2021

Seattle, WA

- **Video Understanding:** Led video understanding research and delivered production models for compliance, tracking, and segmentation within Rekognition Video and related AWS services, used by Prime Video and Amazon Stores.
- **Open-source:** Contributed to GlueCV-Torch, GlueMM, and Decord, providing widely used and flexible computer vision and transformer-based implementations to the community.
- **Research:** Led research in video and multimodal understanding with publications in CVPR, ICCV, ECCV, InterSpeech, and NeurIPS.

EDUCATION

Ph.D. in Computer Engineering, Rutgers University

Sep 2013 - Feb 2018

B.S. in CE, Uni. of Electronic Science and Technology of China

Sep 2009 - June 2013

SKILLS

M-LLM Training: Multimodal LLM training up to Hundred-B scales (video/image/audio); 3D parallelism, Megatron/NeMo, large-scale data and streaming pipelines.

Multimodal Understanding: Visual/audio encoders, cross-modal alignment, representation learning and cross-modal reasoning.

Leadership: Led teams of scientists and engineers to drive model design, large-scale training execution, evaluation, and cross-organizational research initiatives.

SELECTED PUBLICATION

Selected publications, full list can be found at Google Scholar. * Denotes equally contributed.

Amazon Nova Models

1. Core Contributor. “Amazon Nova Multimodal Embeddings: Technical report and model card.”
2. Core Contributor. “Amazon Nova Premier: Technical report and model card.”
3. Core Contributor. “The Amazon Nova family of models: Technical report and model card.”

Vision Language Models

1. Yuxiao Chen, Jue Wang, Jingru Yi, Xinyu Li, Davide Modolo. “Learning Compact Video Representations for Efficient Long-form Video Understanding in LMMs” WACV 2026.
2. Wenjin Zhang, Xinyu Li, Chenyang Gao, Ivan Marsic. “SemiVisBooster: Boosting Semi-Supervised Learning for Fine-Grained Classification through Pseudo-Label Semantic Guidance” ICCV 2025.
3. Seon-Ho Lee, Jue Wang, Zhikang Zhang, David Fan, Xinyu Li. “Video Token Merging for Long Video Understanding.” NeurIPS 2024.
4. David Fan, Jue Wang, Shuai Liao, Zhikang Zhang, Vimal Bhat, Xinyu Li. “Text-Guided Video Masked Autoencoder.” ECCV 2024.
5. Chen, Yuxiao, Jianbo Yuan, Yu Tian, Shijie Geng, Xinyu Li, Ding Zhou, Dimitris N. Metaxas, and Hongxia Yang. “Revisiting multimodal representation in contrastive learning: from patch and token embeddings to finite discrete tokens.” CVPR 2023.

Video and MM understanding

1. Jiaojiao Zhao*, Yanyi Zhang*, Xinyu Li*, Hao Chen, Shuai Bing, Mingze Xu, Chunhui Liu, Kaustav Kundu, Yuanjun Xiong, Davide Modolo, Ivan Marsic, Cees G.M. Snoek, Joseph Tighe. “TubeR: Tubelet Transformer for Video Action Detection.” CVPR 2022 **Oral**.
2. A S M Iftekhar, Hao Chen, Kaustav Kundu, Xinyu Li, Joseph Tighe, Davide Modolo. “What to look at and where: Semantic and Spatial Refined Transformer for detecting human-object interactions.” CVPR 2022 **Oral**.
3. Feng Cheng, Mingze Xu, Yuanjun Xiong, Hao Chen, Xinyu Li, Wei Li, Wei Xia. “Stochastic Backpropagation: A Memory Efficient Strategy for Training Video Models.” CVPR 2022 **Oral**.
4. Xinyu Li*, Yanyi Zhang*, Chunhui Liu, Bing Shuai, Yi Zhu, Biagio Brattoli, Hao Chen, Ivan Marsic, and Joseph Tighe. “VidTr: Video Transformer Without Convolutions.” ICCV 2021.
5. Li, Xinyu, Bing Shuai, and Joseph Tighe. “Directional temporal modeling for action recognition.” ECCV 2020. **Spotlight**

OPEN-SOURCE PACKAGES

GluonCV, with 5.9K stars, provides implementations of state-of-the-art (SOTA) deep learning algorithms in computer vision. I was the core contributor on GluonCV pytorch version. (Link)

GluonCV-Transformer, an open-sourced library based on PyTorch, providing a list of SOTA transformer-based research implementations on various image video and multimodal tasks. (Link)